23055022861046

Code: 1504/R

Faculty of Science

BCA I-Year CBCS I-Semester Regular Examinations Dec- 2023

Paper: EFFECTIVE COMMUNICATION

Time: 3 Hours **Max. Marks: 70**

SECTION – A

I. Answer any Five questions. All questions carry equal Marks **(5 x 4 = 20 Marks)**

1. Use the suitable **articles** in the blanks:
   a. Breakfast is served at six in ....morning. (a/ an/ the)
   b. Newyork is ....large city. (a/ an/ the)
   c. She is ....best friend of mine. (a/ an/ the)
   d. This is ....unusual experience. (a/ an/ the)
2. Verb Forms: **Tenses**
   a. He was ....by a mad dog.(bite)
   b. She ....milk everyday.(drink)
   c. Our visitors ....yesterday.(arrive)
   d. She ....a huge bungalow. (have)
3. Choose the correct option:
   a. Sumathi ....a wise girl .(is /are/was)
   b. We ....a big National.(are/was/were)
   c. There ....no need of your help .(is/are/been)
   d. India ....a big country.(Is/are/be)
4. Write suitable **prepositions** in the blanks:
   a. He met Krishna ....a super market.
   b. we work ....Monday to Friday.
   c. My father bought a new bike ....me.
   d. She goes to college ....bus.
5. Identify the **pronouns** from the sentences:
   a. She is the same height as .... sister.
   b. I bought the coffee with...........money.
   c. You can't blame ....for everything.
   d. ....are trying to flog their house.
6. Write Homonyms for the following words.
   a. Buy e. Sell
   b. Deer f. Die
   c. Flower g. Hair
   d. Knot h. Piece
7. Correct the FOUR sentences that are **Misplaced modifiers:**
   a. Tom barely skidded six inches in the milk spill.
   b. The pedestrian was hot by a car, sitting on the curb.
   c. claiming a neck injury the bungee cord company was used by a jumper.
   d. Contaminated with oil from the spill, Jeff refused to purchased the property.
8. Write the expressions for the following one word substitution.
   a. Cannibal
   b. bilingual
   c. Ecology
   d. Optimist

SECTION B

II. Answer any ALL questions. All questions carry equal Marks (5 x 10 = 50 Marks)

9. A). i. What is the difference between written and oral communication ? 5 M

ii. Why is Knapp's relationship model important? 5 M

OR

B). i. Read carefully and answer the questions from the **Unseen Passage** : 5 M

Taking selfies is one of the most popular activities of today's time. Selfie means to capture a photo of yourself using your smartphone. When the craze of taking selfies increased in the people, a product called Selfie Stick was very much sold in the market.

By setting the smartphone in the selfie stick, people could easily take a selfie of many people in a single frame. But as time passed, the craze of selfie sticks ended.

Now, people use only their smartphones to take selfies and share them on social media platforms like Facebook, Instagram, Twitter, Snapchat, etc.

With the advancement, the kinds of selfies have also changed. Now, you can take a 3D selfie on your smartphone. The difference between 2D and 3D selfies is that 2D selfies can be viewed only from the angle from which it's taken, whereas 3D selfies can be viewed from any angle.

Questions –

a. What is meant by a selfie?
b. Why did people use selfie sticks?
c. Explain the difference between 2D and 3D selfies.
d. Why do people take selfies?
e. Write the antonym of the words 'Advancement'.

ii. what is team work and why does it matter ? 5 M

10. A). i. What is the difference between verbal and Nonverbal communication? 5 M

ii. Read carefully and answer the questions from the **Unseen Passage** 5 M

In the olden times, people were dependent on three things: food, clothes, and house. They used to do agriculture for food. Whatever they grew in the fields, they fed themselves and their family. Cotton and hand-made woolen clothes were the most popular clothes of that time. People used to wear cotton clothes in summer and hand-made woolen clothes in winter.

Since earlier most of the people lived in the villages, so their houses were of mud. Due to not much-advanced technology, they used only limited resources for living.

But today, the time has completely changed. Today's people prefer to live in cities rather than villages. They like living in luxurious houses and bungalows, traveling to new places, eating different types of ***cuisines*** in hotels and restaurants, using new advanced technology, etc.

Before, people lived comfortably with limited resources, but today they have a lot of stress.

Questions –

a. How was the life of the people in the olden times?
b. What things did people depend on in the olden times?
c. Write the synonym of the word 'Cuisine'.
d. What kinds of clothes were the most popular?
e. Where do people like to live in t

OR

B). i. What is meant by emotional quotient? 5 M

ii. A Comprehension Passage: 5 M

Avul Pakir Jainulabdeen Abdul Kalam was an Indian aerospace scientist who served as the 11th President of India from 2002 to 2007. He was born and raised in Rameswaram, Tamil Nadu, and studied Physics and aerospace engineering.

The birth anniversary of Dr. APJ Abdul Kalam is observed as **World Students' Day** which was declared by the United Nations Organisation in 2010.

The "Missile Man of India" and "People's President of India" are the nicknames of the former President of India and Indian scientist Dr. A.P.J. Abdul Kalam. His association with India's Space and Missile Development Program gave him the identity of "Missile Man of India". Take this quiz on Dr. A.P.J. Abdul Kalam to know more about him.

Dr. APJ Abdul Kalam death anniversary is observed every year on July 27. He passed away in 2015 while giving a lecture at the Indian Institute of Management, Shillong at the age of 83 after suffering from cardiac arrest.

a. What is the full name of Dr. Abdul Kalam?
b. When was Dr. A.P.J. Abdul Kalam born?
c. Which island is named after Dr. A.P.J. Abdul Kalam?
d. Dr. A.P.J. Abdul Kalam was the ......President of India.
e. What is the birthplace of Dr. APJ Abdul Kalam?

11. A). i. Write Tenses and their structure with one example each 5 M

ii. Write Verb forms for the following words 5 M

a) Awake b) Drink c) Speak d) Write e) Go

OR

B). i. What are the steps of report writing explain? 5 M

ii. Write a letter to the principal for leave of absence for sister marriage. 5 M

12. A). i. Write an essay on Health is wealth? 5 M

ii. Antonyms 5 M

a) Bravery b) Artificial c) Conceal d) Expensive e) Immense

OR

B). i. Write root word and prefix for the following words. 5 M

a) Immature b) inactive c) preschool d) Restore e) ungrateful

ii. Write meaning for the following phrasal verbs . 5 M

a) Hang on b) Break up c) call off d) give in e) look into

13. A). i. What is meant by email etiquette and why it is important? 5 M

ii. Synonyms : 5 M

a) Amazing b) dead c) independent d) huge e) abundant

OR

B). i. What is the importance of 'Johari window' in personal change management? 5 M

ii. Discuss the dos and don'ts of precise writing? 5 M

******

230550228611046.

Code: 1503/R

Faculty of Science

BCA I-Year CBCS I-Semester Regular Examinations Jan- 2024

Paper: DIGITAL PRINCIPALS

Time: 3 Hours

Max. Marks: 70

SECTION – A

I. Answer any Five questions. All questions carry equal Marks **(5 x 4 = 20 Marks)**

1. i) $(4096)_{10} = (?)_8$ ii) $(1654)_8 = (\ ? \ )_{16}$
2. Explain 4x1 multiplexer with diagrams
3. Explain clocked RS – flipflop with truth table
4. Explain Johnson counter with diagram.
5. Draw any one circuit with latch in A synchronous sequential circuits
6. Explain 2 to 4 decoder with truth table.
7. Explain 1 bit comparator and draw the logic diagram.
8. Subtract $(11010110)_2-(10110010)_2$ using 2's complement method.

SECTION B

II. Answer any ALL questions. All questions carry equal Marks **(5 x 10 = 50 Marks)**

9. A) Explain Basic logic gates and Universal gates with symbols and truth tables.

OR

B)Convert the following $\bar{A}+\bar{B}$ to Canonical form and Write associative laws and Demorgan's theorems

10. A) Simplify f= πM (0,1,2,4,6,8,10,11,12) using K-Map and implement the circuit using logic gates.

OR

B) Simplify Y= (A,B,C,D) = $\sum m$ (0,1,3,7,8,9,11,15) using tabular method.

11. A) Explain JK-flipflop using truth table and circuit diagram.

OR

B) Explain state reduction and assignment design procedure with one example.

12. A) Explain 4 bit synchronous counter with truth table and timing diagrams.

OR

B) Define Register and Explain types of shift registers with neat diagrams.

13. A) Explain design procedure for asynchronous sequential circuits.

OR

B) Explain Analysis procedure in asynchronous sequential circuits.

*****

23055022861l046

Code: 1505/R

Faculty of Science

BCA I-Year CBCS I-Semester Regular Examinations Jan - 2024

Paper: MATHEMATICAL FOUNDATIONS OF COMPUTER SCIENCE

Time: 3 Hours

Max. Marks: 70

SECTION – A

I. Answer any Five questions. All questions carry equal Marks (5 x 4 = 20 Marks)

1. Define the following laws of logic (i) De Morgan's Laws (ii) Commutative Laws.
2. For $\mathcal{U} = \{1,2,3, \dots \dots 9,10\}$. Let $A = \{1,2,3,4,5\}$, $B = \{1,2,4,8\}$, $C = \{1,2,3,5,7\}$ and $D = \{2,4,6,8\}$ then determine $\bar{C} \cup \bar{D}$.
3. Define function, domain, co domain and range of a function.
4. Determine the sequence generated by the generating function $f(x) = (2x - 3)^3$.
5. Define directed graph and subgraph.
6. Define Group.
7. If A= {1,2,3} and B= {2,3} then find A×B and B×A.
8. Define semi group and Monoid.

SECTION B

II. Answer any ALL questions. All questions carry equal Marks (5 x 10 = 50 Marks)

9. A) State and prove Division Algorithm.

OR

B) (i) Use the laws of set theory, simplify the expression $\overline{\overline{(A \cup B) \cap C} \cup \bar{B}}$.

(ii) Define Tautology. Construct a truth table for compound statement $(p \wedge q) \rightarrow p$.

10. A) (i) The functions $f, g, h: \mathbb{R} \rightarrow \mathbb{R}$ are defined such that $f(x) = x^2$, $g(x) = x + 5$ and $h(x) = \sqrt{x^2 + 2}$ then find $(hog)of$ and $j\,o(goh)$.

(ii) Draw the Hasse diagram for the poset $\mathcal{P}(\mathcal{U}, \subseteq)$ where $\mathcal{U} = \{1,2,3\}$.

OR

B) Define equivalence relation. If $A = \{1,2,3,4\}$, give an example of a relation $\mathscr{R}$ on A which is (i) reflexive and symmetric but not transitive

(ii) reflexive and transitive but not symmetric

11. (a) (i) Find the unique solution for the recurrence relation $a_{n+1} - 1.5a_n = 0, n \geq 0$.

(ii) Find the generating function for the sequence 1,-1,1,-1,…

OR

B) Solve the recurrence relation $2a_{n+3} = a_{n+2} + 2a_{n+1} - a_n, n \geq 0$, $a_0 = 0, a_1 = 1, a_2 = 2$.

12. A) Prove that the set G=$\{1, \omega, \omega^2\}$ (cube roots of unity) is an Abelian group with respect to Multiplication.

OR

B) Prove that the set G={0,1,2,3} forms an Abelian group with respect to addition modulo 4.

13. A) Define (i) Euler circuit (ii) regular graph (iii) degree of vertex in a graph (iv) connected graph (v) complete graph

OR

B) Determine |V| for the following graph G

(i) G has nine edges and all vertices have degree 3

(ii) G has ten edges with two vertices of degree 4 and all others of degree 3

*****

230550228611046

Code: 1501/R

Faculty of Science

BCA I-Year CBCS I-Semester Regular Examinations Jan- 2024

Paper: Programming in C

Time: 3 Hours | Max. Marks: 70

SECTION – A

I. Answer any FIVE of the following questions. (5 x 4 = 20 Marks)

1. Define a Computer. Specify different Computer Languages.
2. What are the Type Qualifiers? Describe.
3. What are Pre processor commands?
4. Write the Concepts of Strings in C.
5. Write a short note on Streams.
6. Convert 257 decimal number to Binary and 10001 binary number to decimal.
7. Explain "continue" statement with example program.
8. Write about Command Line Arguments.

SECTION B

II. Answer the following questions. (5 x 10 = 50 Marks)

9. A) Write the Structure of Simple C program. Write the applications of C programs. Specify the Tokens used in C.

OR

B) What are the different Operators used in C. Write their Precedence. Explain Expression evaluation in C.

10. A) Differentiate between if,else-if and switch statements. Give examples to each.

OR

B) Define Functions. What are user defined Functions? Write a program to explain the usage of them.

11. A) Define an Array? Write the applications of Arrays. Write a C program to find the 2x2 matrix Addition.

OR

B) What is searching? Write a C program for Binary Search. And trace the program with sample values.

12. A) What are Pointers? Write the uses of Pointers. Explain Array of Pointers with example program.

OR

B) Describe String Manipulation Functions in C.

13. A) Define a Structure. How to initialize and access Structures? Explain Self Referential Structures.

OR

B) What are Files? Discuss the Modes of Files.

*****

2305502286110026

Code: 1502/BL

Faculty of Science

BCA I-Year CBCS I-Semester Regular Examinations June- 2024

Paper: Web Technologies

Time: 3 Hours

Max. Marks: 70

SECTION – A

I. Answer any Five questions. All questions carry equal Marks (5 x 4 = 20 Marks)

1. How is the BOM used? Explain properties and methods.
2. What are the new input types available in HTML5?How can you create custom input fields with HTML5?
3. What are the different group selectors available in CSS3?Provide examples of using group selectors for common tasks like selecting child elements or descendant elements.
4. How can you define the background properties of an element in CSS3?What are the different types of background properties available?
5. What is the CSS Grid Layout and how can it be used to create responsive layouts?
6. What are the key benefits and challenges of implementing RWD?
7. What is JavaScript and what are its key features?
8. How can you use JSX to create dynamic and interactive web applications?

SECTION B

II. Answer any ALL questions. All questions carry equal Marks (5 x 10 = 50 Marks)

9. A) i. What is a web server and what is its function?How do web servers store and deliver web pages?What are some of the popular web server software options?
ii. What is a web browser and what is its function?What are some of the popular web browsers available today?What are the key features of a web browser?

OR

B) i. What is the difference between id and class attributes?When should you use an id and class attributes?
ii. What is the difference between inline and block elements?How do inline and block elements affect the layout of a web page?What are some examples of inline and block elements?

10. A) i. How does CSS3 improve the design and user experience of web pages?What are the key benefits of using CSS3?
ii. Explain the advantages and disadvantages of each cssmethod:Inline styles, Embedded styles, External stylesheets .

OR

B) i. Explain the concept of the CSS box model and how it defines the space around an element.
ii. How can you control the visibility of elements using the visibility property?How can you use the display property to hide, show, or change the display type of elements?

11. A) i. What are the essential principles and techniques for creating responsive websites?
ii. How can you optimize image file sizes for faster loading and improved mobile performance?

ii. What are the different techniques for delivering responsive images? How can you optimize image file sizes for faster loading and improved mobile performance?

OR

B) i. What is the basic syntax for a media query and what are the different types of media queries available?

ii. How can you use media types to create different styles for different output devices?

12. A) i. How do you interact with HTML forms using JavaScript?

ii. What are the different data types available in JavaScript

OR

B) i. What are the different methods and properties available for working with strings?

ii. What are the different types of events available in JavaScript? Explain.

13. A) i. What are interfaces in TypeScript and how do they define the structure of objects?

ii. How can you define and use functions in TypeScript with type annotations for arguments and return values?

OR

B) i. What are the different types of modules available in TypeScript? Explain.

ii. What is JSX (JavaScript XML) and how can it be used to write React components in TypeScript?

*****